

Spurious Correlations in Machine Learning: A Survey

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Abstract

Machine learning systems are known to be sensitive to *spurious correlations* between biased features of the inputs (e.g., background, texture, and secondary objects) and the corresponding labels. These features and their correlations with the labels are known as “spurious” because they tend to change with shifts in real-world data distributions, which can negatively impact the model’s generalization and robustness. In this survey, we provide a comprehensive review of this issue, along with a taxonomy of current state-of-the-art methods for addressing spurious correlations in machine learning models. Additionally, we summarize existing datasets, benchmarks, and metrics to aid future research. The paper concludes with a discussion of the recent advancements and future research challenges in this field, aiming to provide valuable insights for researchers in the related domains.

1 Introduction

While machine learning (ML) systems have achieved remarkable success, distribution shifts and adversarial examples have become critical issues [Taori *et al.*, 2020]. It is becoming clearer that one of the major causes behind these issues is the high reliance on spurious correlations between biased features of the inputs (e.g., background, texture, and secondary objects) and the corresponding labels. When the correlations captured during training no longer hold in the test data, the performance of ML models tends to deteriorate, resulting in robustness issues and negative social impact in critical domains, such as healthcare. For example, previous trials in recognizing pneumonia with Convolutional Neural Networks (CNNs) show that these models rely on the metal tokens in the chest X-ray scans from different hospitals, instead of learning the pathological features of the disease [Kirichenko *et al.*, 2023]. Therefore, it is critical to directly address the impact of spurious correlations.

In recent years, spurious correlations have been studied under various names, such as shortcuts, dataset biases, group robustness, simplicity bias, and so on. We have seen significant progress in analyzing and mitigating spurious correlations in various areas such as computer vision [Wang *et al.*, 2021],

natural language processing [Du *et al.*, 2022b], and healthcare [Huang *et al.*, 2022]. Despite the progress, there lacks a survey in this area that formally defines spurious correlations and comprehensively introduces and summarizes learning algorithms and other related problems to provide insights into future research.

In this paper, we present the first survey on spurious correlations with their formal definition, the taxonomy of current state-of-the-art methods for addressing spurious correlations in machine learning models, as well as datasets, benchmarks, and metrics, to aid future research. We also discuss the future research challenges and the role of foundation models in this field. We hope that this survey can provide a comprehensive review for interested researchers and also provide valuable insights into future research in related domains.

2 Spurious Correlations

Spurious correlation, namely “*correlations that do not imply causation*” in statistics, refers to a situation where two variables appear to be related to each other, but the relationship is coincidental or confounded by an external variable. This phenomenon leads to misleading or incorrect interpretations of data and models. We first give a formal definition below.

Definition 1 (Spurious Correlation). Let $\mathcal{D}_{tr} = \{(x_i, y_i)\}_{i=1}^n$ be the training set with $x_i \in \mathcal{X}$ and $y_i \in \mathcal{Y}$, where $\mathcal{X}$ denotes the set of all possible inputs, $\mathcal{Y}$ denotes the set of K classes. For each data point x_i with class label y_i , there is a spurious attribute $a_i \in \mathcal{A}$ of x_i , where a_i is non-predictive of y_i , and $\mathcal{A}$ denotes the set of all possible spurious attributes. A spurious correlation, denoted as $\langle y, a \rangle$, is the association between $y \in \mathcal{Y}$ and $a \in \mathcal{A}$, where y and a exist in a one-to-many mapping $\phi : \mathcal{A} \mapsto \mathcal{Y}^{K'}$ conditioned on $\mathcal{D}_{tr}$ with $1 < K' \leq K$. A set of data with the spurious correlation $\langle y, a \rangle$ is annotated with the group label $g = (y, a) \in \mathcal{G}$, where $\mathcal{G} := \mathcal{Y} \times \mathcal{A}$ is the set of combinations of class labels and spurious attributes.

For example, in the training set $\mathcal{D}_{tr}$, we may find two spurious correlations, $\langle y, a \rangle$ and $\langle y', a \rangle$, i.e., a exists in two classes of images and can be mapped to two class labels. A model trained on $\mathcal{D}_{tr}$ may simply use a to predict y , but this will result in incorrect predictions on samples with $\langle y', a \rangle$.

2.1 Where do Spurious Correlations Come from?

Spurious correlations often stem from selection biases on datasets, which consist of several common issues. Datasets

with limited data are often underspecified, where multiple plausible hypotheses can describe the data equally well. For machine learning models, only one of the simplest hypotheses will be preferred according to Occam’s Razor principle. The preferred choice can be suboptimal and incorrectly learn the easier spurious patterns in the data. Imbalanced group labels can result in certain majority groups being over-represented, leading the model to rely on spurious correlations that do not always hold for minority groups. This reliance can cause the model to fail when the data distribution is shifted in real-world applications. Sampling noise in the dataset, which refers to the random variation inherent in data samples collected from a larger population, can also cause spurious correlations. The sampling noise can lead the dataset not to reflect the real data distribution, causing the model to erroneously interpret the noise as a meaningful correlation and make predictions based on that.

2.2 Why are Machine Learning Models Sensitive to Spurious Correlations?

Inductive Biases from Learning Algorithms

The “no free lunch” theorem in machine learning highlights that no single model is universally superior, emphasizing the importance of inductive bias – the set of assumptions an algorithm uses for predictions on new data. These biases are critical to the algorithm’s design and affect its performance. For example, CNNs are biased towards local connectivity and spatial invariance, suitable for image data, while Transformers have no strong inductive biases and are more flexible but more data-hungry. This role of inductive bias becomes especially crucial when dealing with biased datasets. When a model’s inductive bias aligns closely with spurious patterns in the training data, it can exacerbate the problem of spurious correlation. This occurs when the model incorrectly assumes these irrelevant patterns to be predictive, leading to overfitting and poor generalization to new data. Such models fail to capture the underlying, more generalizable patterns while performing well on the biased training data. This issue highlights the need for careful consideration of both the inductive biases of the model and the representativeness of the training data to mitigate the risk of learning misleading correlations.

Optimization

Formally, in the classical Empirical Risk Minimization (ERM) setting [Vapnik, 1991], let the triplet (x, y, a) be a sample x with its label y and a spurious attribute a which is typically unknown. Assume the group $g = (y, a)$ and (x, y, a) follow the distributions P_g and $P_{x|g}$ conditioned on $\mathcal{D}_{tr}$. Given a model with parameters θ as $f_\theta : \mathcal{X} \rightarrow \mathbb{R}^K$ and a loss function $\ell : \mathbb{R}^K \times \mathcal{Y} \mapsto \mathbb{R}$, e.g., softmax cross-entropy, the objective is to minimize the training loss $\mathcal{L}_{avg}$:

$$\mathcal{L}_{avg}(f_\theta) := \mathbb{E}_{g \sim P_g} \mathbb{E}_{(x, y, a) \sim P_{x|g}} [\ell(f_\theta(x), y)]. \quad (1)$$

Since the optimization process does not take spurious attribute a into consideration, when y and a exhibit a strong correlation, the ERM trained model will prioritize a from $f_\theta(x)$ for predicting the label y if a is a relatively easier feature to learn. Upon testing on a dataset where this correlation is absent, the classifier’s reliance on the spurious attribute a as a predictive

feature can lead to incorrect predictions. It has been proved that machine learning models are optimized via ERM in Eq. 1 and have poor worst-group test errors [Zhang *et al.*, 2022] in Eq. 2.

$$\mathcal{L}_{wg}(f_\theta) := \max_{g \in \mathcal{G}} \mathbb{E}_{(x, y, a) \sim P_{x|g}} [\ell(f_\theta(x), y)] \quad (2)$$

To achieve low worst-group test error, prior work Group DRO [Sagawa *et al.*, 2019] proposed modified objectives that focus on the worst-group loss which optimizes for the worst-group training loss or reweighting. However, there remains a strong assumption that group information ($\mathcal{G}$) is available during training. Without such information, different gradient descent algorithms may converge to different local minima. For example, gradient descent with large learning rates tends to learning simple patterns that are shared across examples, while it facilitates complex pattern learning and memorization with small learning rates [Geirhos *et al.*, 2020].

2.3 Related Areas

This subsection offers an overview of research areas closely linked to spurious correlations, providing broader context and complementary views. Domain generalization represents a higher-level concept overlooking the challenges of spurious correlations. Adjacent fields like Group Robustness, Shortcut Learning, and Simplicity Bias intersect with our focus area, sharing some overlapping principles.

Domain Generalization. Domain generalization (DG), also known as out-of-distribution (OOD) generalization, aims to solve a problem where a model is trained on one or more related domains with the intent of generalizing to new, unseen test domains [Wang *et al.*, 2023]. DG is a broad area of study and it varies depending on the assumptions made about the relationship between training and test distributions. A significant threat to domain generalization is spurious correlations, where features that are predictive in the training distribution no longer correlate with the label in the test distribution [Yang *et al.*, 2022].

Invariant Learning. The concept of invariant learning (IL) is creating models that utilize patterns or features in data that remain invariant across various domains [Arjovsky *et al.*, 2019; Chen *et al.*, 2022]. This is beneficial when training data originates from diverse sources, yet some fundamental principles or relationships are consistent. To accomplish invariant learning, it is crucial to train models in a way that they do not rely on spurious correlations.

Group Robustness. Group robustness refers to the consistency and fairness of a machine learning model’s performance across different groups or subpopulations within the data [Liu *et al.*, 2021; LaBonte *et al.*, 2023]. If a model depends on correlations unique to a specific group (for example, a correlation found in training data that predominantly represents a certain demographic), it may not perform well for other groups where these correlations don’t apply. One way to mitigate spurious correlation is from the group robustness perspective, aiming to maximize the accuracy of the worst-performing groups (usually the minority groups) while maintaining high average accuracy [Yang *et al.*, 2022].

Shortcut Learning. Shortcuts refer to spurious correlations that models learn, which perform well in standard bench-

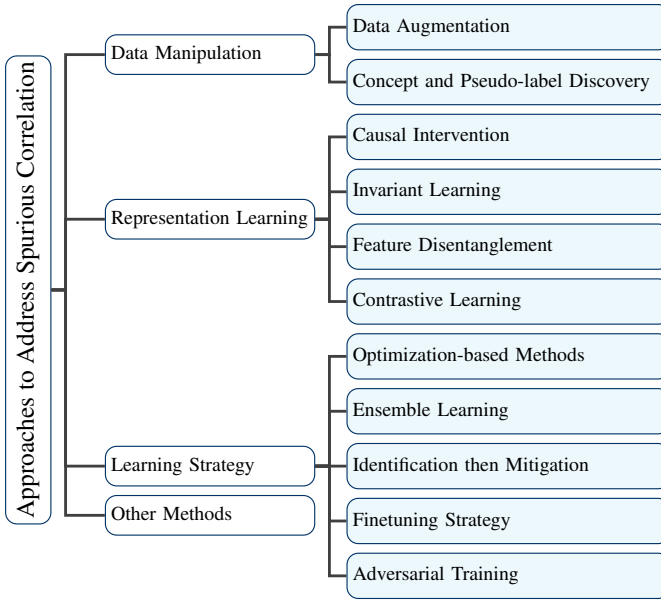


Figure 1: A comprehensive taxonomy of approaches to address spurious correlation in machine learning.

mark environments but struggle in more complex testing domains [Geirhos *et al.*, 2020; Du *et al.*, 2022a]. This phenomenon, known as shortcut learning, often occurs when models are trained under empirical risk minimization (ERM). Such models find it difficult to generalize to relevant or challenging distribution shifts [Yang *et al.*, 2022].

Simplicity Bias. Simplicity bias (SB) is the inclination of deep neural networks to favor less complex patterns (such as the background) over more intricate ones (like the object’s semantics), even if all features have the same predictive power [Tiwari and Shenoy, 2023]. This leads to vulnerabilities to minor adversarial perturbations (due to the small margin) and spurious correlations (with simpler features) [Shah *et al.*, 2020].

3 Methods

In this section, we will discuss how various machine learning techniques address the issue of spurious correlation. These techniques work at various stages in the pipeline of machine learning model training, which we summarize into the following categories: data manipulation, representation learning, learning strategy, and other methods. Figure 1 shows the overall taxonomy of the methods to be surveyed.

Data Manipulation. This group includes methods that modify the input of the learning algorithm to enhance the generality and diversity of data distribution. Such modifications may involve augmenting new data samples, generating auxiliary group information in the existing dataset, or other strategies to balance or diversify the data.

Representation Learning. This category focuses on learning improved representations within the models. Various perspectives on the factors that might affect the model’s robustness will be explored, including *causal intervention*, *feature disentanglement*, *invariant learning*, and *contrastive learning*. These techniques aim to help the model better understand the

underlying relationships between variables and become more resilient to spurious correlations.

Learning Strategy. This group focuses on designing strategies that can be employed during the learning process to enhance the model’s generalization capability. This may include *optimization*, *ensemble learning*, *identification then mitigation*, *finetuning strategy*, or *adversarial training* that can help prevent strong reliance on spurious correlations and improve the model’s ability to generalize to new data.

Other Methods. This category includes methods that are specific to certain problem settings, such as multi-task problems, reinforcement learning, or Test-Time Adaptation problems. These methods may not be applicable to all situations, but they offer valuable insights and solutions for addressing spurious correlations in their respective domains.

3.1 Data Manipulation

Data Augmentation

Data augmentation is a technique used to increase the diversity of a dataset without collecting new data. This is done by making adjustments to the training data, such as image rotation, cropping, or noise injection. The goal of data augmentation is to improve the generalization capabilities of machine learning models by reducing the impact of spurious correlations present in the original data.

In some cases, data augmentation makes use of additional annotated data to leverage humans’ common sense knowledge of causality. One example of this is UV-DRO [Srivastava *et al.*, 2020], which uses human annotation to augment training examples with potential unmeasured variables. This reduces the spurious correlation problem to a covariate shift problem.

When additional annotation is not available, the following methods use existing training data to augment the dataset. Nuisances via Negativa [Puli *et al.*, 2022] proposes a data augmentation method to corrupt semantic information to produce models that can identify and adjust for where nuisances drive predictions. LISA [Yao *et al.*, 2022] is a simple mixup-based technique that learns invariant predictors via selective augmentation. This selectively interpolates samples either with the same labels but different domains or with the same domain but different labels. The Counterfactual Generator [Zeng *et al.*, 2020] creates counterfactual examples by making interventions on existing observational examples. This is used to enhance the original dataset and improve the performance of machine learning models.

Concept and Pseudo-label Discovery

Another line of work in data manipulation involves predefining concepts and generating pseudo-labels to enhance the conceptual information before training. This process helps break intrinsic spurious correlations within the training data and improve the model’s robustness. Discover and Cure (DISC) [Wu *et al.*, 2023] is a method that discovers unstable predefined concepts across different environments as spurious attributes. It then augments on the training data using the discovered concepts to reduce spurious correlations. Spread Spurious Attribute (SSA) [Nam *et al.*, 2022] is an example of generating pseudo-labels. It uses a small amount of spurious attribute annotated samples to estimate group information for the training

set samples. The generated pseudo-labeled attribute together with the training set is then used to train a robust model using worst-case loss minimization.

3.2 Representation Learning

Causal Intervention

Causal Intervention methods focus on the causal relationship between input, label, and any potential spurious attributes. These methods aim to improve model robustness and fairness by minimizing the impact of spurious attributes.

Several methods consider the causal intervention as an implicit design inside the model structure. [Wang *et al.*, 2021] introduce a Causal Attention Module (CaaM) that utilizes unsupervised learning within the Vision Transformer architecture to self-annotate and mitigate confounding effects. By doing so, CaaM enhances model performance and promotes causality. Causal VQA [Agarwal *et al.*, 2020] uses a semantic editing-based approach to assess the robustness of Visual Question Answering (VQA) models. The model demonstrates significantly less flipping behavior under invariant and covariant semantic edits, resulting in improved causality. Counterfactual invariance measures how the predicted label behaves under interventions on portions of input data.

Other methods consider formal ways of adding structured causal graphs to reflect the true causal relationship between the attributes inside the data. [Veitch *et al.*, 2021] examines a true causal structure and applies causal inference tools to formalize and study perturbative stress tests. Latent Causal Invariance Models (LaCIM) [Sun *et al.*, 2021] represents a set of causal models that account for causal structure, aiding in the identification of causal relationships for predictions. De-confound-TDE [Tang *et al.*, 2020] offers a principled solution to pinpoint the causal effect of momentum in long-tailed classification tasks using causal graphs.

Invariant Learning

Invariant learning focuses on creating models that can identify and leverage invariant relationships in data. This approach is particularly crucial and well-noticed in addressing the issue of spurious correlations, which occurs when a model mistakenly relies on non-causal or irrelevant features that happen to correlate with the target variable in the training data, but do not generalize well to new data.

Earlier work IRM [Arjovsky *et al.*, 2019] proposed an objective to learn deep, invariant features and correlations of data which are a complex function of latent variables, which has been a principled method in general domain generalization problems. It identifies and leverages features with consistent predictive performance, avoiding reliance on spurious correlations and ensuring stability across various domains.

Several later works follow the path of IRM and try to emphasize different aspects of the invariant feature. From the risk perspective, REx [Krueger *et al.*, 2021] focuses on model performance across varied risk scenarios. By training models under different risk levels, REx encourages models to avoid overfitting to specific features of the training environment.

From the environment perspective in the image data, EIIL [Creager *et al.*, 2021] aims to learn invariant features in the

various environments of the images. It automates the inference of different environments within datasets and performs well especially when explicit labels of the environments are unavailable. SFB [Eastwood *et al.*, 2023] selectively amplifies features stable across environments, minimizing dependence on non-generalizable, spurious features, aiming to enhance the stability of features in the model.

A more theoretical work [Chen *et al.*, 2022] analyzes the necessary conditions for group invariant learning for spurious-free predictors, and obtains two group criteria, i.e., falsity exposure and label balance. This paper thus proposes SCILL to satisfy the two criteria which is provably sufficient in solving spurious correlations.

Feature Disentanglement

Feature Disentanglement method in machine learning aims to separate spurious and general representations within the latent space. These disentanglements are often achieved by a dual/multiple-branch design in the model structure. StableNet [Zhang *et al.*, 2021] gets rid of both linear and non-linear dependencies between features by utilizing the characteristics of Random Fourier Features (RFF), together with another standard classifier branch to compose up the final loss. [Lee *et al.*, 2021] trains two distinct encoders to separately extract bias features and intrinsic features. They further enhance model robustness by augmenting bias-conflicting samples through the random swapping of vectors between these encoders. ChromaVAE [Yang *et al.*, 2022] utilizes a dual-pronged Variational Auto-Encoder (VAE) to disentangle latent representations and learn shortcut features in a smaller latent subspace. This method is focused on ensuring that these shortcut features do not dominate the learning process.

Contrastive Learning

Contrastive learning has emerged as a promising approach in machine learning, particularly effective in addressing spurious correlations. This method was common in the self-supervised learning paradigm which can learn generalized representations. A notable implementation of this approach is Correct-n-Contrast (CnC) [Zhang *et al.*, 2022]. CnC begins by training an ERM model to identify samples within the same class but with dissimilar spurious features. Following this, it employs a robust model trained through contrastive learning. This model is designed to learn similar representations for same-class samples, despite the presence of varying spurious features. The end result is a more generalizable and resilient model, capable of distinguishing between essential and non-essential features, thus mitigating the influence of spurious correlations.

3.3 Learning Strategy

Optimization-based Methods

In tackling spurious correlations, a range of optimization-based methods have been developed. Optimization methods focus on proposing reasonable training objectives for the models to learn more reasonable metrics, thus improving the model's robustness and fairness. Group Distributionally Robust Optimization (Group DRO) [Sagawa *et al.*, 2019] is the very first work, which focuses on minimizing the worst-case loss across different predefined groups in a dataset. This method is effective in ensuring that the model performs well across various

subpopulations, reducing the risk of overfitting to dominant trends. Conditional Value at Risk (CVaR) DRO [Levy *et al.*, 2020] extends this concept by optimizing the expected loss over the worst-case distribution of the data. This approach is useful for managing risk in scenarios where extreme cases can have significant impacts, thereby promoting a more robust model performance.

Instead of following the worst-case loss from Group DRO, [Liu *et al.*, 2023] proposes a new logit correction (LC) loss by adjusting the model’s output logits based on group membership, aiming to balance performance across different groups. This correction is particularly useful in scenarios where certain groups are underrepresented or more prone to misclassification due to biased data.

Ensemble Learning

Ensemble learning denotes a set of methods that ensembles multiple independently trained biased models to make a more generalizable unbiased model. For instance, the LWBC method, as described by [Kim *et al.*, 2022], utilizes a committee of classifiers to identify bias-conflicting data—those free from spurious correlation—and emphasizes these during the training of the main classifier. Similarly, [Bahng *et al.*, 2020] introduced ReBias, a framework for training de-biased representations by making them statistically independent from a set of intentionally biased representations.

Several methods also consider a two-stage ensemble strategy. [Lee *et al.*, 2023] proposed DivDis, a two-stage framework that initially generates a variety of hypotheses using unlabeled data and subsequently selects the most suitable one with minimal additional supervision. Additionally, the Learning from Failure (LfF) [Nam *et al.*, 2020] approach involves a two-step process where the first network is intentionally trained with bias, and the second network is debiased by focusing on samples that counteract this bias. Collectively, these methods exemplify the diverse strategies in ensemble learning aimed at mitigating spurious correlations in machine learning models.

Identification then Mitigation

There are quite a few spurious correlation methods following an “Identification then Mitigation” strategy. These methods first focus on identifying instances of spurious correlations, then followed by well-designed interventions to mitigate their impact.

The following methods consider identifying the underperforming samples in the dataset and then mitigating the spurious correlations based on previously identified data. Just Train Twice (JTT) [Liu *et al.*, 2021] proposes a simple yet effective strategy. It initially identifies examples that are hard to classify, then re-trains the model to focus specifically on these challenging tasks by upweighting the identified points. First Amplify Correlations and Then Slice (FACTS) [Yenamandra *et al.*, 2023] initially amplifies correlations to fit a simplified bias-aligned hypothesis through empirical risk minimization. Then it applies correlation-aware slicing, using mixture modeling in the bias-aligned feature space to identify and address underperforming data slices representing distinct correlations. SePArate early and REsample (SPARE) [Yang *et al.*, 2023a]

operates by distinguishing large groups with spurious correlations early in the training process. It then employs importance sampling to balance these group sizes, effectively alleviating the influence of spurious correlations.

Other methods aim to quantify spurious correlations as scores in the dataset before taking corresponding mitigative actions. Less-Learn-Shortcut (LLS) [Du *et al.*, 2023] focuses on quantifying the degree of bias in samples and proportionately down-weighting them. This method allows for a more balanced learning process by reducing the influence of heavily biased data. Spuriousity Rankings [Moayeri *et al.*, 2023] involves sorting data within each class according to their level of spuriousity, as determined by neural features in an interpretable network. This technique facilitates the efficient retrieval of natural examples and counterfactuals related to relevant spurious cues from large datasets, thereby enabling effective measurement and mitigation of reliance on spurious features.

Additionally, there is also one method [Yadav *et al.*, 2022] which transforms the natural language data to a rule-based logic formulation and then uses logical negation to address spurious correlations. The proposed Tsetlin Machine (TM) employs a rule-based architecture that learns both simple and complex correlations by combining features and their negations. This approach enables the extraction of rule-based reasoning from TM, yielding explainable AND-rules that incorporate both negated and non-negated elements.

Overall, these methods highlight a divide-and-conquer approach to identifying and mitigating spurious correlations, effectively enhancing the reliability and fairness of machine learning models.

Finetuning Strategy

Finetuning strategies focus on refining existing general prediction models by selectively adjusting either a portion of the model or the data used. This selective finetuning often targets outlier data and crucial interpretable parts of the model, aiming to address the challenge of spurious correlations. MaskTune [Asgari *et al.*, 2022] encourages a model to consider a broader range of input variables, mapping them to the same target, thereby promoting a more comprehensive understanding of the data. Influence Tuning [Han and Tsvelkov, 2021] takes a unique approach by backpropagating a target instance’s attribution interpretation directly into the model, allowing for more targeted adjustments. Explanation-Based Finetuning [Ludan *et al.*, 2023] involves training the model on artificially constructed datasets that include various types of spurious cues, and subsequently testing it on sets devoid of these cues.

Recently, some methods only consider a relatively simple strategy, last-layer finetuning, while they achieve surprisingly good performance. Deep Feature Reweighting (DFR) [Kirichenko *et al.*, 2023] focuses on retraining the last layer of a network to reduce the impact of spurious features, enhancing the model’s worst-group performance and reducing reliance on non-essential features, like background and texture. Selective Last-Layer Finetuning (SELF) [LaBonte *et al.*, 2023] employs a strategy of finetuning the last layer using a small, more group-balanced dataset, showing improved performance to near-DFR levels with no group annotations.

Dataset	#Class	#Sample	Brief Description	Domain
Colored MNIST	-	60,000	Synthetic Dataset	Vision
Corrupted CIFAR-10	-	60,000	Synthetic Dataset	Vision
CelebA	2×2	202,599	Face Attribute	Vision
Waterbirds	2×2	11,788	Bird Subset of CUB	Vision
ImageNet-9	9	5,495	Subset of ImageNet	Vision
ImageNet-A	200	7,500	Subset of ImageNet	Vision
ImageNet-C	75	-	Subset of ImageNet	Vision
BAR	6	2,595	Action	Vision
NICO	19	24,214	Animal and Vehicle	Vision
MetaShift	410	12,868	Natural Images	Vision
FMOw	63	1,047,691	Satellite Images	Vision
Spawrious	6×4	152,064	Dog	Vision
CivilComments	7	1,999,514	Public Comment	Text
MultiNLI	3×2	206,175	Language Inference	Text
FDCL18	7	80,000	Abusive Behavior	Text
IV-VQA	-	657,090	Causal VQA	VQA
CV-VQA	-	59,513	Causal VQA	VQA
ChestX-ray14	14	112,120	Chest X-Ray	Health
MIMIC-CXR	-	377,110	Chest X-Ray	Health
CheXpert	14	224,316	Chest X-Ray	Health
PadChest	19	160,868	Chest X-Ray	Health

Table 1: Datasets for spurious correlation in machine learning.

Adversarial Training

Adversarial training strengthens a model against adversarial inputs, enhancing its ability to generalize and learn on general data patterns, thereby reducing the effects of spurious correlations. Backdoor Poisoning Attacks [He *et al.*, 2023] is an adversarial attack method where the attacker injects several watermarked, mislabeled training examples into a training set. This paper argues that backdoor poisoning attacks exhibit spurious correlations between simple text features and classification labels, and accordingly proposes methods for mitigating spurious correlation as a means of defense.

3.4 Other Methods

This category collects the methods with unique problem settings. While these techniques may not be universally applicable, they provide insightful solutions for mitigating spurious correlations within their specialized fields. Test-Time Label-Shift Adaptation (TTLA) [Sun *et al.*, 2023] addresses label shifts occurring at test time. MultiTask Causal Representation Learning (MT-CRL) [Hu *et al.*, 2022] enhances learning in multi-task environments by identifying causal relationships. RSC-MDP [Ding *et al.*, 2023] offers a novel perspective in reinforcement learning settings, focusing on reducing spurious dependencies for more reliable decision-making processes. Another work [Yang *et al.*, 2023b] also focuses on applying multi-modal models to leverage different modalities to explicitly set apart spurious attributes from the affected class.

4 Datasets and Metrics

4.1 Datasets

In this subsection, we briefly introduce the popular datasets for spurious correlation in machine learning (Table 1). Under the spurious correlation settings, the correlation between the label y and the spurious attribute a at training time does not hold at test time. Most of benchmarks are constructed from these existing datasets. Vision tasks are the most common used benchmarks to understand spurious correlations,

such as Colored MNIST [Lee *et al.*, 2021], Corrupted CIFAR-10 [Lee *et al.*, 2021], CelebA [Sagawa *et al.*, 2019], Waterbirds [Sagawa *et al.*, 2019], subset of ImageNet [Hendrycks and Dietterich, 2018; Xiao *et al.*, 2020; Hendrycks *et al.*, 2021], FMOW [Christie *et al.*, 2018], BAR [Nam *et al.*, 2020], NICO [He *et al.*, 2021], MetaShift [Liang and Zou, 2021], Spawrious [Lynch *et al.*, 2023], and 2 VQA benchmarks [Agarwal *et al.*, 2020]. In NLP, CivilComments [Borkan *et al.*, 2019], MultiNLI [Sagawa *et al.*, 2019], and FDCL18 [Founta *et al.*, 2018] are also widely used benchmarks. Apart from vision and language areas, spurious correlation is also an important research topic for AI in healthcare. For instance, ChestX-ray14 [Wang *et al.*, 2017], CheXpert [Irvin *et al.*, 2019], PadChest [Bustos *et al.*, 2020], MIMIC CXR [Johnson *et al.*, 2019] are widely used chest X-ray datasets to understand spurious correlations in medical imaging.

4.2 Metrics

The most common metric to evaluate the robustness of a model f_θ to spurious correlations is the worst-group-accuracy $\text{Acc}_{\text{wg}}(f_\theta)$ [Idrissi *et al.*, 2022; Chaudhuri *et al.*, 2023], which is defined as the minimum accuracy across groups of test samples, i.e.,

$$\text{Acc}_{\text{wg}}(f_\theta) := \min_{g \in \mathcal{G}} \mathbb{E}_{(x,y) \sim \mathcal{D}_{\text{test}}^g} [\mathbb{1}(f_\theta(x) \neq y)], \quad (3)$$

where $g = (y, a)$ denotes the group label determined by the target y and the spurious attribute a , $\mathcal{G}$ denotes the set of all possible group labels, (x, y) denotes an image-label pair, $\mathcal{D}_{\text{test}}^g$ denotes the set of image-label pairs that have the spurious attribute a and target y which determine the group label g , and $\mathbb{1}(\cdot)$ denotes the identity function which gives one if the input condition is true and give zero otherwise. If the model does not exploit spurious attributes, the worst-group-accuracy will be high on test sets $\mathcal{D}_{\text{test}}^g$.

Except for the worst-group-accuracy, there may exist other evaluation protocols such as the average of all group accuracy $\text{Acc}_{\text{ag}}(f_\theta)$ define as

$$\text{Acc}_{\text{ag}}(f_\theta) := \mathbb{E}_{g \in \mathcal{G}} \mathbb{E}_{(x,y) \sim \mathcal{D}_{\text{test}}^g} [\mathbb{1}(f_\theta(x) \neq y)], \quad (4)$$

and bias-conflicting accuracy $\text{Acc}_{\text{bc}}(f_\theta)$ defined as,

$$\text{Acc}_{\text{bc}}(f_\theta) := \mathbb{E}_{g \in \mathcal{G}_{\text{bc}}} \mathbb{E}_{(x,y) \sim \mathcal{D}_{\text{test}}^g} [\mathbb{1}(f_\theta(x) \neq y)], \quad (5)$$

where $\mathcal{G}_{\text{bc}}$ denotes labels of pre-specified bias-conflicting groups. Typically, these bias-conflicting group labels are the labels of the minority groups in the training set. The regular average accuracy over all test samples is also used when the test set is specifically constructed for evaluating the robustness to spurious correlations.

All these metrics can be used together or separately depending on the availability of group labels and the types of test datasets. These metrics provide holistic views on whether the model can eliminate correlations between classes and spurious attributes.

5 Discussion

5.1 Summary

Spurious correlations are first tackled under group robustness which optimizes the worst-group performance of a model

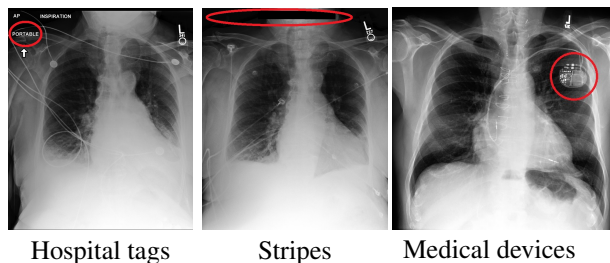


Figure 2: Hospital tags, strips, and medical devices exemplify several unknown group labels in the MIMIC-CXR dataset, which might spuriously correlate with the ground truth diagnosis results.

given that group labels in data are known. However, knowing group labels is a strong assumption in practice as it often requires expert knowledge and human annotations. Many recent works adopt various learning and group-label-inference strategies aiming to relax the requirement. However, they still require a small portion of held-out data that are annotated with group labels for model selection or training group label estimators. Therefore, completely removing the requirement for group labels is important to pave the way for wider applications of spurious correlation mitigation methods, such as medical area (In Figure. 2).

Many methods targeting at spurious correlations optimize the worst-group accuracy which is widely used as a measure of robustness to spurious correlations. However, in practice, there is an irreconcilable tradeoff between the worst-group and average performance of a model; many methods achieve higher worst-group accuracy at the cost of reduced average accuracy. This begs the question: is the metric a good optimization objective? Moreover, spurious correlations often arise from the context around a target, e.g., grass field provides a common context for predicting the target cow. Therefore, it is worth exploring whether eliminating spurious correlations is a good strategy overall.

5.2 Future Research Challenges

Based on the above discussions, we summarize several promising research challenges below.

Group-Label-Free Spurious Correlation Mitigation. While existing methods greatly relax the requirement of group labels, they still need a certain amount of group labels to formulate their learning objectives, which is still a significant barrier for their wide applications. Developing more realistic spurious correlation mitigation methods without the need of group labels is a promising future direction.

Automated Spurious Correlation Detection. Mitigating the impact of spurious correlations requires us to first identify them. Group labels provide a very limited number of spurious correlations in data. Human annotations are often adopted to identify diverse spurious correlations. However, this is not a scalable approach. Moreover, this may miss some spurious correlations that do not align with human understandings, such as low-level patterns not perceivable by humans. Developing automated spurious correlation detection methods may greatly advance the development in this field.

Balancing the worst-and-average performance tradeoff. Existing methods improve a model’s robustness to spurious

correlations, which is typically measured in worst-group accuracy, at the cost of decreased average accuracy. Such an unfavorable tradeoff makes it difficult to choose which models to be deployed in practice. Spurious correlations may provide useful contexts for predictions in some scenarios, and focusing exclusively on mitigating the reliance on spurious correlations may not be an optimal strategy. Therefore, it is worth exploring more effective methods to balance this tradeoff.

Rigorous Evaluation Benchmarks. There is a scarcity of rigorous evaluation benchmarks for evaluating a model’s robustness to spurious correlations. Group-label-based benchmarks only provide robustness guarantee against a few spurious correlations. Models may develop reliance on other spurious correlations while mitigating the reliance on targeted ones. It is also critical to ensure that different kinds of spurious correlations are balanced so that the evaluation will not be biased towards models that favor specific spurious correlations.

5.3 How will Foundation Models Affect this Area?

The recent surge in generative AI has spotlighted foundation models, which are large deep neural networks and are trained on exceptionally massive data. Due to the enormous amount of training data, these models can perform a wide range of disparate tasks and are able to reflect real-world data distributions. Therefore, one promising direction is to use foundation models as a tool for solving spurious-correlation-related problems. For example, we can design input prompts to instruct foundation models to detect spurious correlations or to selectively generate more diverse data to mitigate the reliance on spurious correlations.

Another promising direction is to study the spurious correlations in foundation models. As foundation models are overparameterized models, a critical concern is that overparameterization in these models may exacerbate spurious correlations [Sagawa *et al.*, 2020]. This suggests that while being powerful, foundation models may also enhance unintended or misleading spurious associations, especially on previously unseen data. The hallucination of Large Language Models (LLMs) is a well-acknowledged example, where LLMs occasionally generate spurious content that diverges from user inputs, contradicts previously generated context, or misaligns with established factual knowledge. New methods, such as introducing knowledge graphs into foundation models, may be developed to address this issue.

6 Conclusion

Spurious correlations represent a ubiquitous challenge in machine learning, particularly as they contribute to model brittleness and poor performance under domain shifts. This survey thoroughly reviews the issue, presenting a detailed taxonomy of current methods, datasets, and metrics designed to evaluate the robustness under spurious correlations. We give a summary of the current stage of this area and suggest several future research challenges. Moving forward, particularly in the era of large foundational models, our discussion seeks to lay the groundwork for future progress in this field. We hope that this survey will inspire further research in addressing spurious correlations and contributing to advancements in the machine learning community.

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